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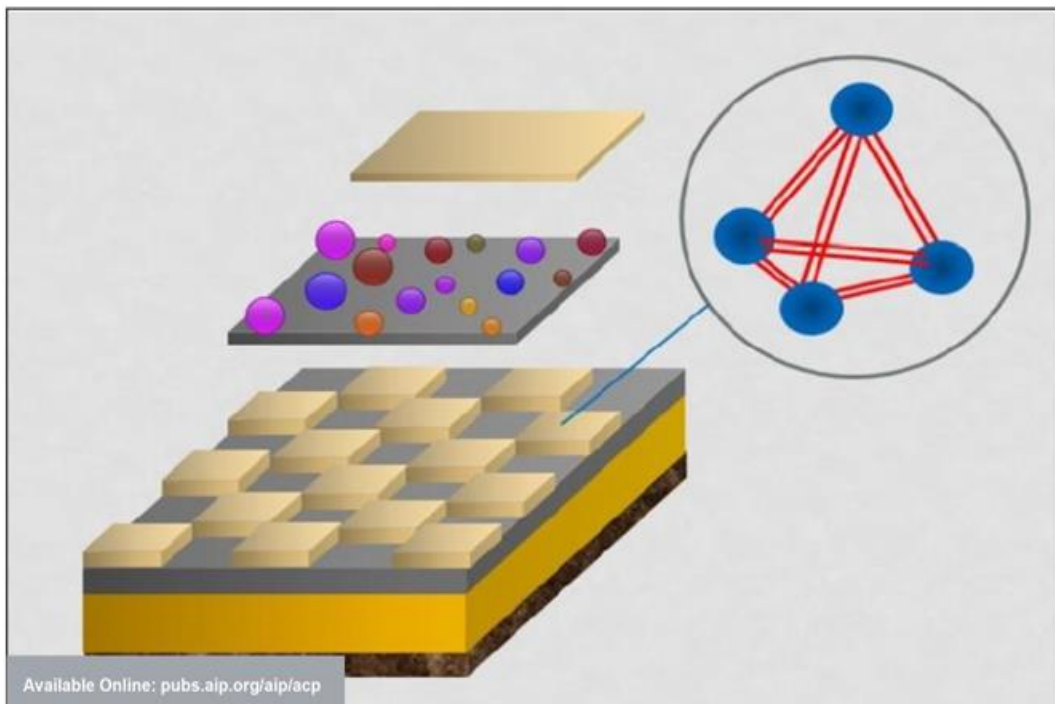


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Study of Sliding Wear and Erosion wear of HVOF sprayed Alloy 718 and Al₂O₃ composite coating on GCI substrate

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Abstract.In this study superalloy Alloy-718 has been selected as a coatings with Al₂O₃ as reinforcement, and four samples were used in accordance with the addition of Al₂O₃ to Alloy718 from 10% to 30% and wear performance and Alloy-718/Al₂O₃ composite coatings' erosion behaviour has been studied to study the mechanical and micro-structural characteristics of super alloys combined with ceramics in the form of composite coatings. GCI has been selected as a substrate. Furthermore, the HVOF thermal spray technique has been selected for the present work as it offers the smoothest, hardest, lowest porosity and most homogenous microstructure coatings that are well bound. On a high-temperature pin-on-disc tribometer with unidirectional dry sliding, tribological studies have been conducted. Testing is conducted in accordance with ASTM G99-04 standard. At room temperature, 400°C, and 800°C, the tests were conducted with two distinct load values of 25 and 50 N, a speed of 1 ms⁻¹, and a sliding distance of 2000 m and it is found that the Abrasive wear co-efficient was calculated by using Archard's Wear Relation and found that Alloy-718+30%Al₂O₃ 1.2X 10E⁻⁵ which is lower than other composite coated samples. Erosive wear rate is 1.9X10⁻⁴ and 1.8X10⁻⁴ at 30degrees and 60 degrees Eroded angles also less when compared to other coated samples.

Keywords –GCI, Super Alloy-718 Matrix and Al₂O₃ as reinforcement, Dry Sliding Wear study and Erosion study.

INTRODUCTION

Super alloy-718 has been selected as a replacement over conventional Ni-Cr coatings with Al₂O₃ as reinforcement, Moreover, limited research has been presented by researchers on the post-heat treatment and wear performance of Alloy-718/ Al₂O₃ composite coatings¹⁻⁴. There is need to study the mechanical and micro-structural characteristics of super alloys combined with ceramics in the form of composite coatings. Super alloys have high creep strength, good oxidation resistance, but due to ductile nature its hardness may be low and same can be improved by adding ceramics⁵⁻⁸. Following analysis of the literature review, the substrate material for the current study was chosen and the past work done and future work that can be conducted and are feasible in the present life situations and may help the society⁹⁻¹². GCI has been selected as a substrate. Furthermore, the HVOF thermal spray technique has been selected for the present work as it offers the smoothest, hardest, lowest porosity and most homogenous microstructure coatings that are well bound¹³⁻¹⁸. Further, the microstructure of coatings before and after the wear and corrosion test would be checked by using characterization techniques like FE-SEM/EDS analysis to ensure the effectiveness of coatings in terms of porosity, density, presence of semi-melted particles, cracks and to understand the failure mechanism of wear and corrosion¹⁹⁻²¹.

EXPERIMENTAL PROCEDURE AND MATERIALS

Erosion and wear test of coated, uncoated and post processed samples in simulated boiler environment at high temperature by using the pin-on-disc and air-jet erosion tester. The GreyCast Iron samples on which composite coatings were deposited by using HVOF thermal spray process in variation of addition of Al₂O₃ have been used for